

Project Description

You need to print CSE231-X in a seven segment display, where X is your group number.

This project will have two parts.

1. Combinational circuit part:

Here you need to design a combinational circuit where you will provide manual input. For your each input, each letter will be shown. You need to do the combinational circuit in all four different ways using logisim.

- a. using normal gates
- b. using universal gates
- c. using Decoder
- d. using MUX.

For hardware implementation, each group will implement only their respective assigned one.

2. Sequential circuit part:

In this part you need to design a synchronous counter. When you will attach this part with your combinational part, all the digits will keep showing one by one and repeat. **You have to do this part in logisim and hardware as well.**

What to Submit:

You need to submit 4 logisim files where in every file both the sequential and combinational parts are there. You need to come with a laptop to show logisim files. You have to bring the circuit. You need to submit a hand written report. In this report, you need to write down all the procedures (ex: truthtable, KMap, eqn etc) for four different processes of combinational circuits and for sequential circuit as well. No extra writing is needed. You need to write what is the benefit of choosing this combinational circuit process (The one is assigned for your group).